

The dominance order on partitions of n is a self-dual lattice

The Last Math Competition, Conjecture 00000002486

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2026-10-02

Abstract

For every $n \geq 0$, the partitions of n ordered by dominance form a lattice: any two partitions have a meet and a join. Conjugation $\lambda \mapsto \lambda'$ is an involution that reverses dominance, so it exchanges meets and joins and the lattice is self-dual. This is a classical theorem of Brylowski. We give a short self-contained proof and a complete Lean 4 formalization on Mathlib's `Nat.Partition`.

1 The conjecture as stated

The original text of conjecture 00000002486 reads:

Definition: The order on partitions: dominance. Conjecture: The dominance partial order on partitions is a lattice structure: joins and meets exist, and the structure closes under conjugation duality. (partition order dominance lattice structure)

2 Reading of the statement

A partition λ of n is written with its parts in weakly decreasing order, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell > 0$, and we set $\lambda_i = 0$ for $i > \ell$. Its partial sums are

$$\Sigma_k(\lambda) = \lambda_1 + \dots + \lambda_k \quad (k \geq 0).$$

The dominance order on the set $\text{Par}(n)$ of partitions of n is $\lambda \trianglelefteq \mu$ iff $\Sigma_k(\lambda) \leq \Sigma_k(\mu)$ for all k . The conjugate partition λ' has parts $\lambda'_j = \#\{i : \lambda_i \geq j\}$ ($1 \leq j \leq \lambda_1$), the column lengths of the Young diagram.

Dominance compares partitions of the same integer n ; this is the standard setting. So the statement is about the poset $(\text{Par}(n), \trianglelefteq)$ for each n . We read its three clauses as follows.

1. *Partial order*: $\trianglelefteq$ is reflexive, transitive and antisymmetric on $\text{Par}(n)$.
2. *Lattice*: any $\lambda, \mu \in \text{Par}(n)$ have a greatest lower bound (meet) $\lambda \wedge \mu$ and a least upper bound (join) $\lambda \vee \mu$.
3. *Conjugation duality*: conjugation maps $\text{Par}(n)$ to itself, is an involution ($\lambda'' = \lambda$) and reverses the order ($\lambda \trianglelefteq \mu \iff \mu' \trianglelefteq \lambda'$). Hence it is an isomorphism from the lattice onto its dual. It exchanges meets and joins: $(\lambda \wedge \mu)' = \lambda' \vee \mu'$ and $(\lambda \vee \mu)' = \lambda' \wedge \mu'$.

3 Result

The conjecture is true. All three clauses hold for every $n \geq 0$.

Theorem 1. *Let $n \geq 0$.*

- (a) *$(\text{Par}(n), \trianglelefteq)$ is a partial order.*
- (b) *Any $\lambda, \mu \in \text{Par}(n)$ have a meet. It is the partition ν with $\Sigma_k(\nu) = \min(\Sigma_k(\lambda), \Sigma_k(\mu))$ for all k .*
- (c) *$\lambda'' = \lambda$, and $\lambda \trianglelefteq \mu \iff \mu' \trianglelefteq \lambda'$.*
- (d) *m is a meet of λ, μ iff m' is a join of λ', μ' . In particular any λ, μ have the join $\lambda \vee \mu = (\lambda' \wedge \mu')'$, and $(\text{Par}(n), \trianglelefteq)$ is a lattice isomorphic to its dual via conjugation.*

4 Proof

Throughout, $\lambda \in \text{Par}(n)$. Since every part is at least 1, there are at most n parts, so $\Sigma_0(\lambda) = 0$ and $\Sigma_k(\lambda) = n$ for $k \geq n$. The increments $\Sigma_{k+1}(\lambda) - \Sigma_k(\lambda) = \lambda_{k+1}$ are weakly decreasing, so

$$\Sigma_k(\lambda) + \Sigma_{k+2}(\lambda) \leq 2\Sigma_{k+1}(\lambda) \quad (k \geq 0). \quad (1)$$

Lemma 2 (Realization). *Let $S : \mathbb{N} \rightarrow \mathbb{N}$ be nondecreasing with $S(0) = 0$, $S(k) + S(k+2) \leq 2S(k+1)$ for all k , and $S(k) = n$ for $k \geq n$. Then there is exactly one $\lambda \in \text{Par}(n)$ with $\Sigma_k(\lambda) = S(k)$ for all k .*

Proof. Put $d_k = S(k+1) - S(k) \geq 0$. The hypothesis says $d_{k+1} \leq d_k$, and $d_n = 0$. Let J be the least index with $d_J = 0$. Then $d_0 \geq \dots \geq d_{J-1} > 0$ and $d_k = 0$ for $k \geq J$. So $d_0 + \dots + d_{J-1} = S(n) - S(0) = n$, and the partition λ with parts $d_0, \dots, d_{J-1}$ satisfies $\Sigma_k(\lambda) = d_0 + \dots + d_{k-1} = S(k)$. For uniqueness, the parts of any partition are recovered from its partial sums as $\lambda_{k+1} = \Sigma_{k+1}(\lambda) - \Sigma_k(\lambda)$. $\square$

(a) Reflexivity and transitivity are inherited from $\leq$ on $\mathbb{N}$. If $\lambda \trianglelefteq \mu \trianglelefteq \lambda$, then $\Sigma(\lambda) = \Sigma(\mu)$, so $\lambda = \mu$ by the uniqueness in Lemma 2.

(b) Let $S(k) = \min(\Sigma_k(\lambda), \Sigma_k(\mu))$. It is nondecreasing, $S(0) = 0$, and $S(k) = n$ for $k \geq n$. For concavity, say $S(k+1) = \Sigma_{k+1}(\lambda)$. Then by (1),

$$S(k) + S(k+2) \leq \Sigma_k(\lambda) + \Sigma_{k+2}(\lambda) \leq 2\Sigma_{k+1}(\lambda) = 2S(k+1).$$

Lemma 2 gives ν with $\Sigma(\nu) = S$. Then $\nu \trianglelefteq \lambda$ and $\nu \trianglelefteq \mu$. If $\rho \trianglelefteq \lambda$ and $\rho \trianglelefteq \mu$, then $\Sigma_k(\rho) \leq S(k) = \Sigma_k(\nu)$, so $\rho \trianglelefteq \nu$.

Conjugation and partial sums. Count the cells of the Young diagram in the first t columns row by row. This gives

$$\Sigma_t(\lambda') = \sum_i \min(\lambda_i, t) = n - G_\lambda(t), \quad G_\lambda(t) := \sum_i (\lambda_i - t)^+. \quad (2)$$

The second equality uses $\min(x, t) + (x - t)^+ = x$.

Lemma 3. For all $J, t \geq 0$ we have $\Sigma_J(\lambda) \leq Jt + G_\lambda(t)$. Equality holds whenever $\lambda_i \geq t$ for $i \leq J$ and $\lambda_i \leq t$ for $i > J$. In particular:

- (i) for every t there is a J with equality, e.g. $J = \#\{i : \lambda_i > t\}$;
- (ii) for every J there is a t with equality, namely $t = \lambda_{J+1}$.

Proof. We have $\Sigma_J(\lambda) = \sum_{i \leq J} \lambda_i \leq \sum_{i \leq J} (t + (\lambda_i - t)^+) \leq Jt + G_\lambda(t)$. Under the stated condition both inequalities are equalities. In (i) and (ii) the condition holds because the parts are weakly decreasing. $\square$

(c), order reversal. Let $\lambda \trianglelefteq \mu$ and $t \geq 0$. Choose J as in Lemma 3(i) for λ . Then

$$G_\lambda(t) = \Sigma_J(\lambda) - Jt \leq \Sigma_J(\mu) - Jt \leq G_\mu(t),$$

and the last step is Lemma 3 for μ . By (2), $\Sigma_t(\mu') \leq \Sigma_t(\lambda')$ for every t , that is, $\mu' \trianglelefteq \lambda'$.

(c), involution. Fix k . By (2) applied to λ' , $\Sigma_k(\lambda'') = n - G_{\lambda'}(k)$, so it suffices to show $\Sigma_k(\lambda) + G_{\lambda'}(k) = n$.

- By Lemma 3(i) for λ' there is J with $\Sigma_J(\lambda') = Jk + G_{\lambda'}(k)$, so $n - G_\lambda(J) = Jk + G_{\lambda'}(k)$ by (2). Lemma 3 for λ gives $\Sigma_k(\lambda) \leq kJ + G_\lambda(J) = n - G_{\lambda'}(k)$.
- By Lemma 3(ii) for λ there is t with $\Sigma_k(\lambda) = kt + G_\lambda(t)$. Lemma 3 for λ' gives $n - G_\lambda(t) = \Sigma_t(\lambda') \leq tk + G_{\lambda'}(k)$, so $\Sigma_k(\lambda) \geq n - G_{\lambda'}(k)$.

Hence $\Sigma(\lambda'') = \Sigma(\lambda)$, and $\lambda'' = \lambda$ by (a). Applying the order reversal to $\mu' \trianglelefteq \lambda'$ gives $\lambda'' \trianglelefteq \mu''$, i.e. $\lambda \trianglelefteq \mu$. This is the converse.

(d) By (c), conjugation is a bijection of $\text{Par}(n)$ onto itself that reverses $\trianglelefteq$ in both directions. Such a map sends greatest lower bounds to least upper bounds and conversely. Applying this to the meet $\lambda' \wedge \mu'$ from (b) shows that $(\lambda' \wedge \mu')'$ is a join of $\lambda'' = \lambda$ and $\mu'' = \mu$. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states and proves Theorem 1 for every n . Partitions are Mathlib's `Nat.Partition n`.

- `sortedParts`, `partialSum`, `Dominates` — the parts in weakly decreasing order, the partial sums Σ_k , and $\trianglelefteq$. `IsMeet`, `IsJoin` — greatest lower and least upper bounds for `Dominates`.
- `collen p j = \#\{i : \lambda_i > j\}` and `conj` — the conjugate, whose parts are the nonzero values among `collen p j`, $0 \leq j < n$.
- `exists_of_concave`, `eq_of_partialSum_eq` — Lemma 2; `exists_meet` — part (b).
- `partialSum_conj` — formula (2).
- `partialSum_le`, `partialSum_split` — Lemma 3.
- `exists_split_threshold`, `exists_split_index` — Lemma 3(i) and (ii).

- `dominates_conj`, `conj_conj`, `dominates_iff_conj` — part (c); `isMeet_iff_isJoin_conj`, `exists_join` — part (d).
- `ConjectureHolds` — for every n : reflexivity, transitivity and antisymmetry of `Dominates`; existence of meets and joins; `conj (conj p) = p`; order reversal; and meets of p, q correspond to joins of their conjugates. `conjecture_00000002486` proves it.
- `dominanceLattice n` packages the result as a Mathlib `Lattice (Nat.Partition n)` whose order is `Dominates`. `conj_sup` and `conj_inf` state $(\lambda \vee \mu)' = \lambda' \wedge \mu'$ and $(\lambda \wedge \mu)' = \lambda' \vee \mu'$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000002486
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The theorem is due to Brylawski [1]. The order reversal under conjugation is also in Macdonald's book [2, Ch. I, (1.11)]. The argument above organizes both through the Legendre-type identity of Lemma 3: $\Sigma_J(\lambda) = \min_t (Jt + G_\lambda(t))$ and $G_\lambda(t) = \max_J (\Sigma_J(\lambda) - Jt)$.

The join is not the pointwise maximum of partial sums. Take $\lambda = (3, 1, 1, 1)$ and $\mu = (2, 2, 2)$ in `Par(6)`. Their partial sums are $(3, 4, 5, 6)$ and $(2, 4, 6, 6)$. The pointwise maximum $(3, 4, 6, 6)$ has increments $3, 1, 2$, which are not decreasing, so it is not the partial-sum sequence of any partition. The join is $(3, 2, 1)$, which equals $(\lambda' \wedge \mu')'$ with $\lambda' = (4, 1, 1)$ and $\mu' = (3, 3)$. The meet is $(2, 2, 1, 1)$, given by the pointwise minimum $(2, 4, 5, 6)$.

References

- [1] T. Brylawski, *The lattice of integer partitions*, Discrete Math. 6 (1973), 201–219.
- [2] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.